# The Beta Distribution

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## Introduction

The beta distribution is the flexible family of probabilities on the unit interval $[0,1]$. It is the natural distribution for proportions, success probabilities, and measurements bounded between two fixed limits. In Bayesian inference, it is the conjugate prior for the binomial and Bernoulli likelihoods.

## Prerequisites

Continuous distributions, Bayesian inference basics.

## Theory

A beta random variable with shape parameters $a > 0$ and $b > 0$ has PDF

$$f(x) = \frac{1}{B(a,b)} x^{a-1}(1-x)^{b-1}, \qquad 0 \le x \le 1,$$

where $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$ is the beta function.

Moments:

- $E[X] = a/(a+b)$.
- $\mathrm{Var}(X) = ab/[(a+b)^2(a+b+1)]$.

Special cases:

- $a = b = 1$: Uniform$(0,1)$.
- $a = b = 0.5$: Jeffreys prior for the binomial probability.
- $a, b > 1$: unimodal density with mode $(a-1)/(a+b-2)$.
- $a < 1$ and $b < 1$: bimodal (poles at 0 and 1).

**Conjugacy**: given a Bernoulli/Binomial likelihood with success probability $\theta$ and a Beta$(a,b)$ prior on $\theta$, the posterior after observing $k$ successes in $n$ trials is Beta$(a+k, b+n-k)$.

## Assumptions

The variable must be bounded between 0 and 1. For probabilities near 0 or 1, the distribution can have a pole; use care when simulating.

## R Implementation

```
curve(dbeta(x, 2, 5),  from = 0, to = 1, col = "steelblue", lwd = 2,
      main = "Beta densities", ylab = "f(x)")
curve(dbeta(x, 5, 2),  add = TRUE, col = "#F4A261", lwd = 2)
curve(dbeta(x, 0.5, 0.5), add = TRUE, col = "#6A4C93", lwd = 2)
curve(dbeta(x, 2, 2),  add = TRUE, col = "#2A9D8F", lwd = 2)
legend("top", legend = c("Beta(2,5)", "Beta(5,2)", "Beta(0.5,0.5)", "Beta(2,2)"),
```

```r
       col = c("steelblue", "#F4A261", "#6A4C93", "#2A9D8F"), lwd = 2)

# Moments
a <- 2; b <- 5
c(mean = a / (a + b), var = a * b / ((a + b)^2 * (a + b + 1)))

X <- rbeta(1e5, a, b)
c(emp_mean = mean(X), emp_var = var(X))

# Bayesian update example
prior_a <- 2; prior_b <- 2
k <- 15; n <- 20
post_a <- prior_a + k; post_b <- prior_b + n - k
c(posterior_mean = post_a / (post_a + post_b),
  posterior_95_CI_lower = qbeta(0.025, post_a, post_b),
  posterior_95_CI_upper = qbeta(0.975, post_a, post_b))
```

## Output & Results

```
   mean      var
 0.2857   0.0255

 emp_mean   emp_var
   0.286     0.025

posterior_mean posterior_95_CI_lower posterior_95_CI_upper
         0.708                 0.504                 0.871
```

Empirical moments match theoretical. The Bayesian update with a Beta(2,2) prior and 15 successes in 20 trials gives a posterior mean of 0.71 with a 95% credible interval of (0.50, 0.87).

## Interpretation

Beta distributions appear everywhere in Bayesian analysis of proportions. Reporting "posterior probability of success 0.71 (95% credible interval 0.50 to 0.87)" is a beta-based statement.

## Practical Tips

- For weakly informative priors, Beta(1, 1) (uniform) or Beta(0.5, 0.5) (Jeffreys) are common defaults.
- Very large $a$ and $b$ correspond to strongly informative priors; check the prior predictive distribution to verify plausibility.
- Beta regression (`betareg`) models proportions in $(0, 1)$ as a function of covariates.
- Values exactly at 0 or 1 are problematic; transform via $(x(n-1) + 0.5)/n$ or use zero-one-inflated beta.
- For a quick-look sanity check, a beta with $a = b$ is symmetric and centred at 0.5, scaling by $(a + b)$ as a concentration parameter.

## For Reviewers

**What to look for in a paper using this method.**

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

## See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots